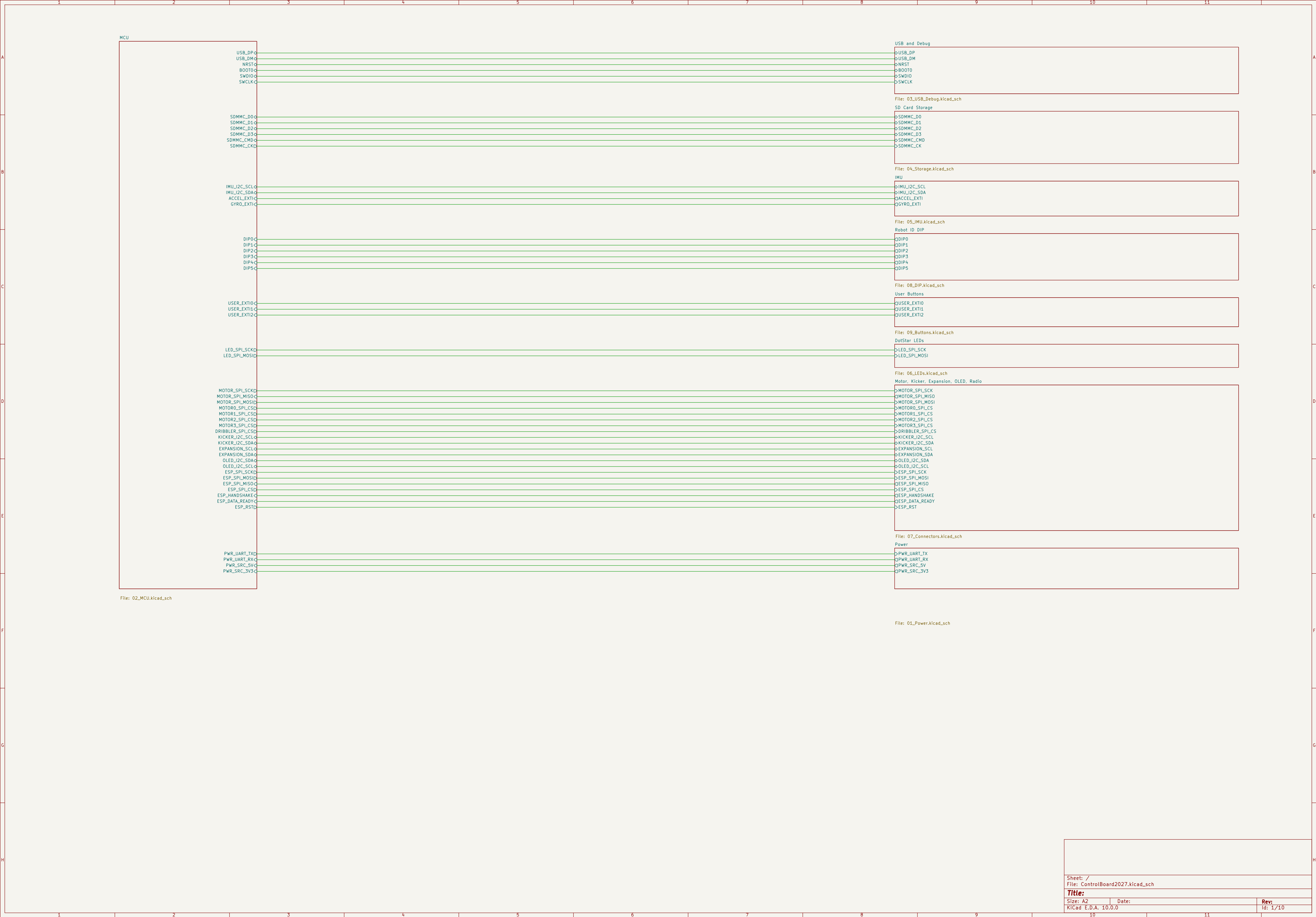
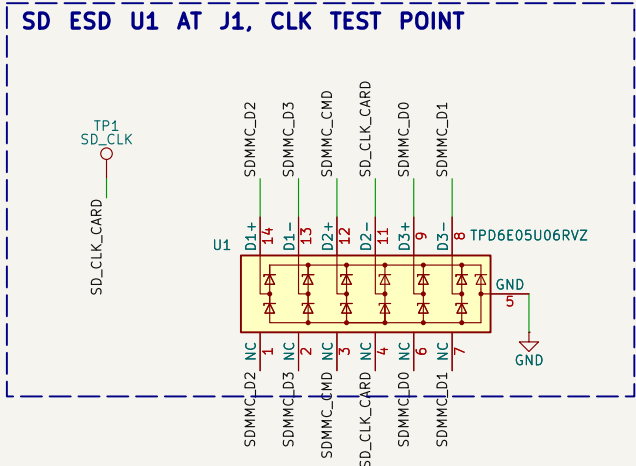
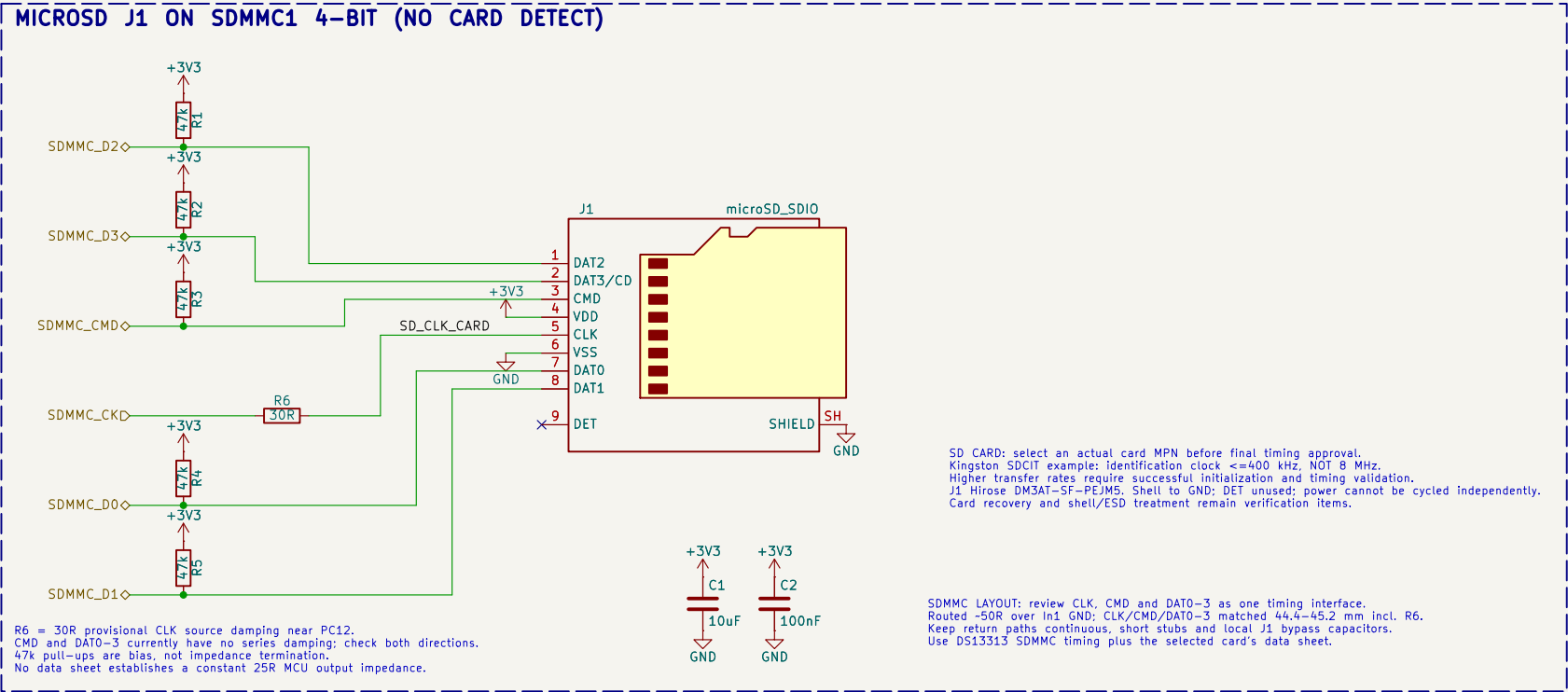
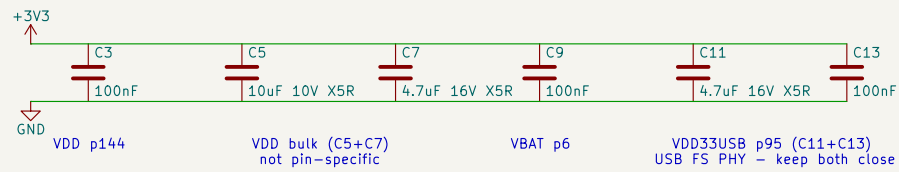
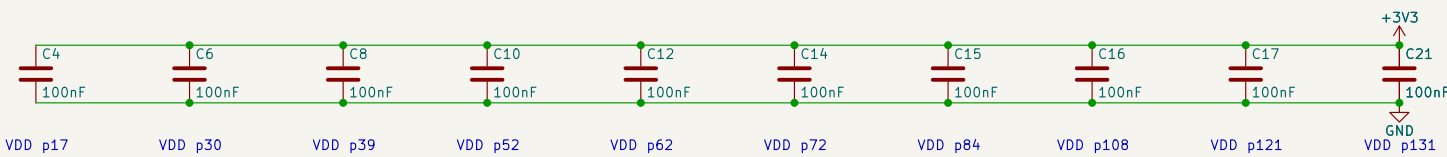






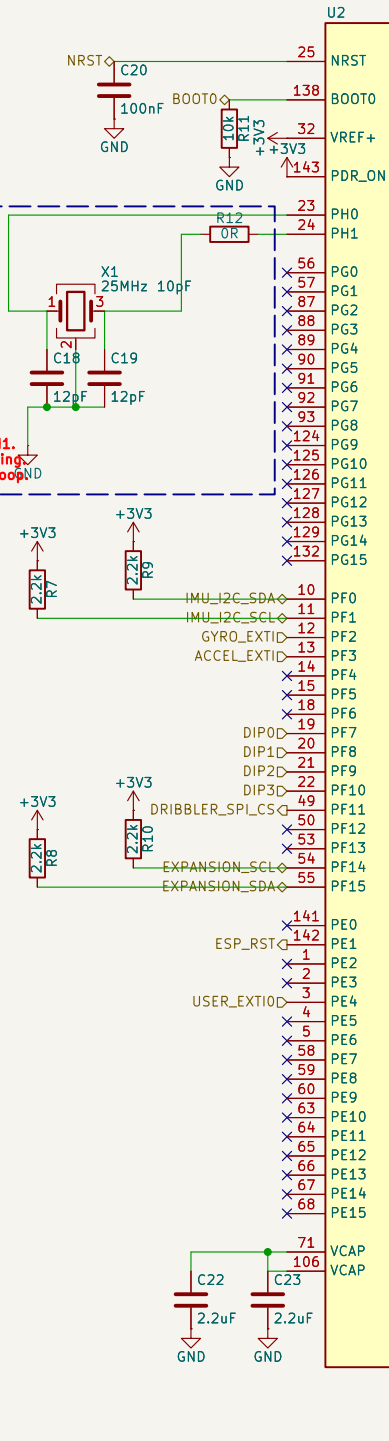
VDD DECOUPLING (EACH CAP AT ITS PIN)

ROUTING – DECOUPLING: each cap beside the STM32 pin named under it, via straight to GND.
C22/C23 (VCAP) tight to p71/p106. The netlist does not keep this pairing.



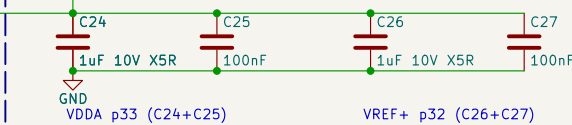
HSE 25MHZ CRYSTAL

ROUTING – HSE CRYSTAL
X1, C18, C19 within 5mm of PH0/PH1.
Short equal OSC traces, GND guard ring.
Nothing routed under or beside the loop.

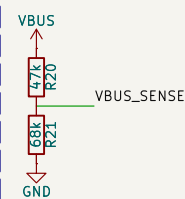


VDDA / VREF+ DECOUPLING

VDDA p33 and VREF+ p32 use +3V3.
Separate local bypass pairs; analog noise must be measured.
Source: DS13313 supply scheme / electrical characteristics.



VBUS SENSE



R15/R16 = 30R; provisional source damping, not certified matching.
No fixed driver resistance or SSR target follows from DS13313 Tables 52/54.
Validate SCK/MOSI/MISO/CS at every cable receiver, both directions.

ROUTING – MOTOR SPI (6 Mbit/s, 5 drops)
R15/R16 within 5mm of PA5/PA7, not at the connectors.

I2C: R7/R9, R8/R10, R14/R17, R18/R19 are each 2.2k.
At 3.3 V: (3.3-0.4)/2.2k = 1.32 mA sink requirement.
For a 300 ns rise-time limit: C <= 300ns/(0.8475R) = 161 pF nominal.
At +5% R: 153 pF; include ESD pins, cables and parallel pull-ups.
LSM6DSGX20X / SSD1306 / DS13313 timing and I/O data; external loads not qualified.

I2C VALIDATION REQUIRED: measure SDA and SCL at worst load.
Do not change to 1.5k from rise time alone; recheck every device's VOL/current.
OLED module pull-ups and expansion-device specifications are not established.

CLOCK: X1 = 25 Mhz; set firmware clock tree to the fitted part.
R12 = 0R is an oscillator drive-tuning link, not line termination.
C18/C19 need measured stray capacitance to establish the 10 pF load.
Startup margin and crystal drive remain unverified (ABM8, DS13313 HSE data).
No blanket claim that 25 Mhz always starts or HSI48 is unusable.
SUPPLIES: PDR_ON and VBAT are on +3V3; no backup battery.
VCAP 71/106 share their bypass node; retained pending ERC symbol review.
Do not separate a required supply node merely to silence ERC.
Source: DS13313 power-supply scheme / VCAP data.

USB-C J2: FUSE, CC RESISTORS, ESD

USB LAYOUT: keep DP/DM coupled over an uninterrupted ground reference.
Join duplicate Type-C data contacts at J2; U3 close to connector.
Final impedance/skew depend on stackup and PHY timing, not series R alone.
DS4260 Figures 6/17: minimize ESD path inductance. Routed 90R pair over In1 GND.

The diagram illustrates the USB-C J2 connector layout and ESD protection circuit. Key components and connections include:

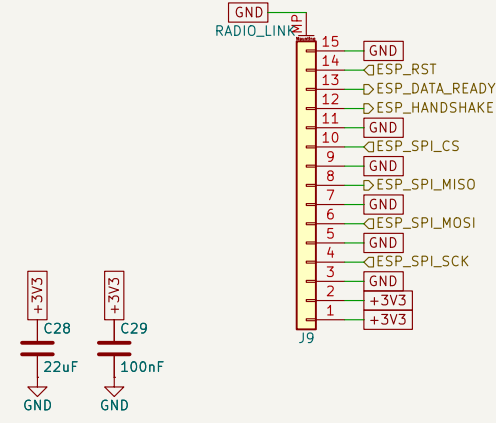
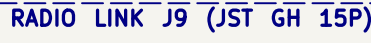
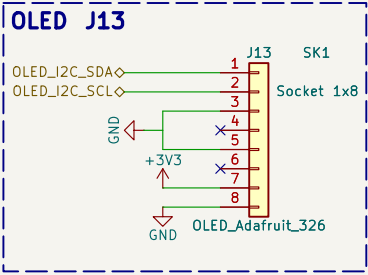
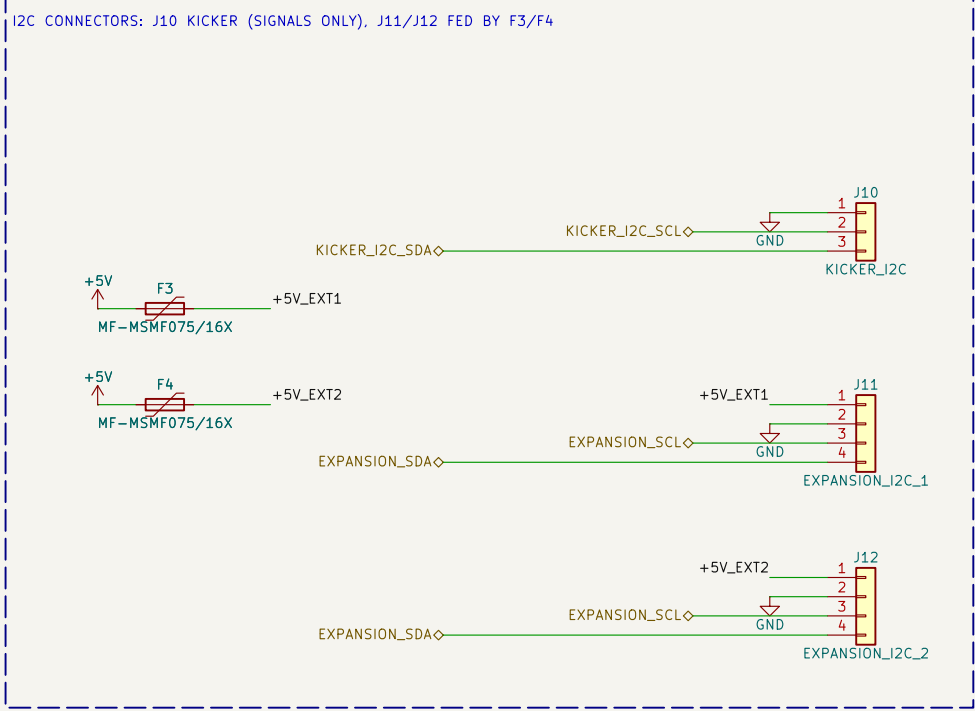
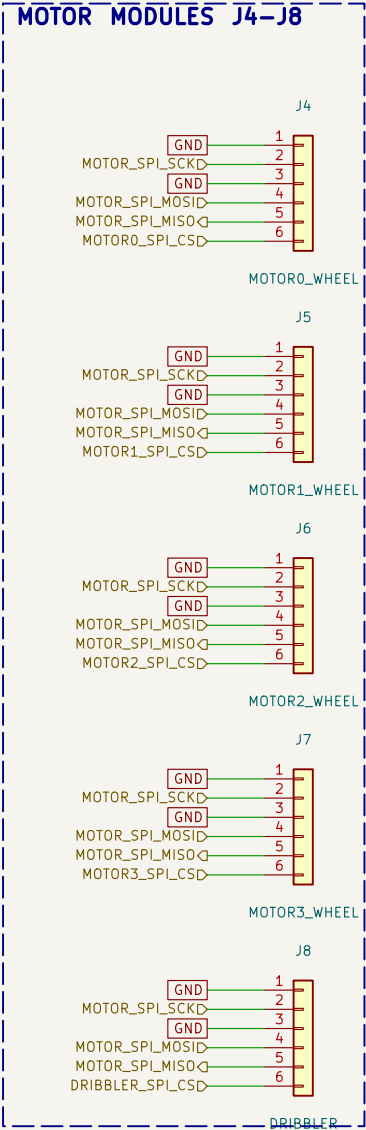
- Connector J2 (USB-C-DFU):** Features pins A4 (VBUS), A5 (CC1), B5 (CC2), A7 (D-), B7 (D-), A6 (D+), B6 (D+), A8 (SBU1), and B8 (SBU2).
- ESD Protection:** A TVS0500DRV (D1) is connected to VBUS and GND. A fuse (F1, MF-MSMF075/16X) is placed on the VBUS line.
- CC Resistors:** 5.1k resistors (R22) are connected between CC1/A5 and CC2/B5 to GND.
- DP/DM Coupling:** A 5.1k resistor (R23) is connected between D+/A6 and D-/B7 to GND.
- Shielding:** A SHIELD is connected to GND via pin A1.
- Power and Ground:** PWR_FLAG and TP3 (TP-VBUS) are connected to VBUS. GND is connected to SH and A1.
- ESD Protection IC (U3):** A USBLC6-25C6 is connected to VBUS (+3V3) and GND. Its pins 1, 2, 3, 4, 5, and 6 are connected to USB_DM_CON and USB_DP_CON.

USB LOAD: MF-MSMF075/16X I_{hold} = 0.75 A at 23C, 0.55 A at 50C.
Actual concurrent worst-case board/external current is not established.
Restricted bench use until host current, startup and thermal budget are checked.
F1 does not negotiate USB current; replacing it alone cannot approve more load.

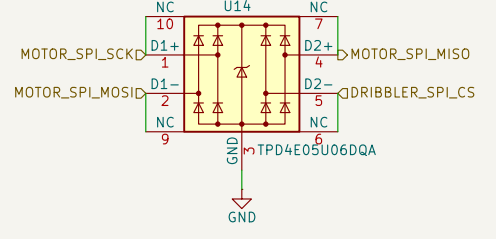
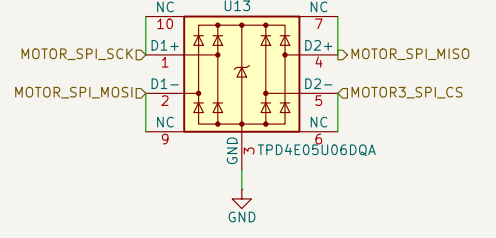
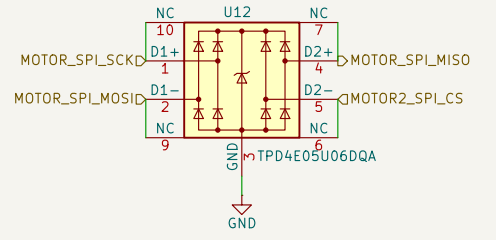
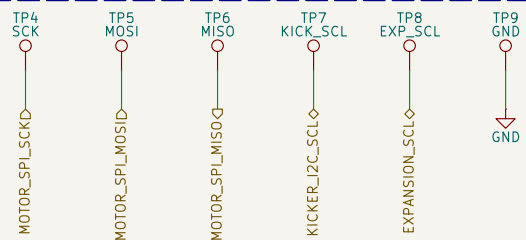
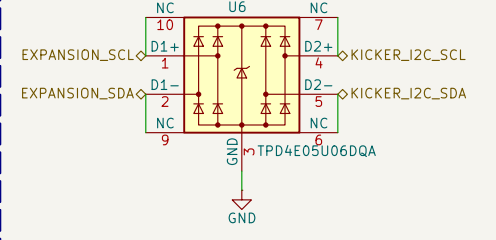
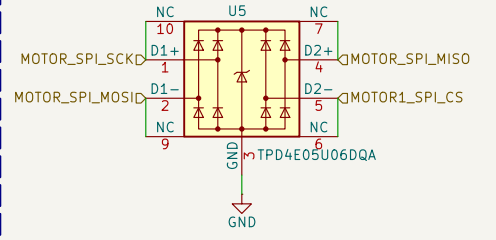
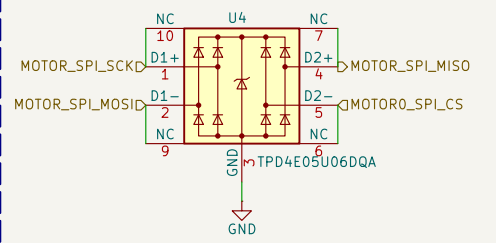
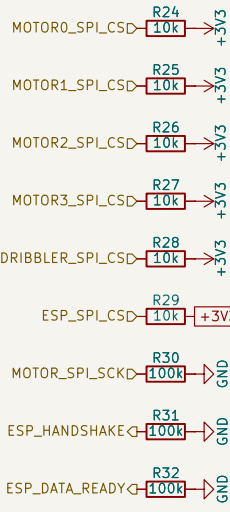
The diagram illustrates the wiring for SWD debug, RESET, and BOOT pins. A 5-pin connector J3 is shown with pins 1 through 5. Pin 1 is connected to +3V3. Pin 2 is connected to SWCLK. Pin 3 is connected to SWDIO. Pin 4 is connected to GND. Pin 5 is connected to NRST. A green line labeled SWD_5pin connects the SWDIO pin to a green line labeled SWD_5pin. A green line labeled NRST is connected to a green line labeled NRST. A green line labeled RESET is connected to a green line labeled RESET. A green line labeled BOOT0 is connected to a green line labeled BOOT0_DFU. A green line labeled +3V3 is connected to a green line labeled +3V3. A green line labeled SW1 is connected to a green line labeled SW1. A green line labeled SW2 is connected to a green line labeled SW2. A green line labeled GND is connected to a green line labeled GND.

SWD is direct; damping depends on probe/cable edges and layout.
 DS13313 does not prescribe a universal 22R or 55R series resistor.
 NRST has an internal 30–50k pull-up; C20 = 100nF (DS13313 NRST data).
 U3 clamp rail is +3V3; validate rail injection and power-off behavior.
 USBLC6–2 DS4260 pinout/topology; short return path required.

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U3 clamp rail is +V3; validate rail injection and power-off behavior.
USBL6-2 DS4260 pinout/topology; short return path required.



J9. CONTROLBOARD INTERFACE ONLY (remote board not audited):
 1/2 +3V3; 3/5/7/9/11/15/MP GND; 4 5CK; 6 MOSI; 8 MISO;
 10 CS; 12 HANDSHAKE; 13 DATA_READY; 14 RESET.
 Confirm cable/mating pinout and remote I/O limits before connection.
 Configure reset open-drain only if the remote EN pull-up is present.
 R31/R32 = 100k: bits when disconnected, not guaranteed reset-state logic.
 ESP32-C5 MTCR/GPIO4 has a reset pull-up; receiver level must be checked.
 Protocol speed/modio and interrupt masking are firmware requirements.

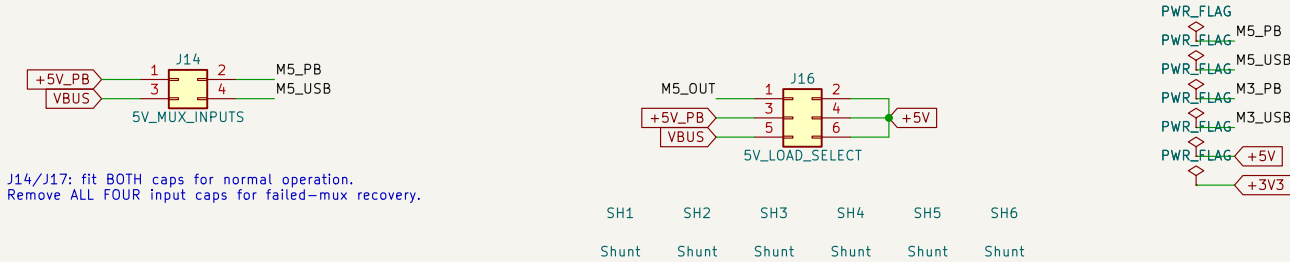


SPI ROUTING: routed ~50R over In1 DND: receiver data still pending.
R15/R16 are provisional damping: verify all multidrop endpoints.
Place ESD devices at connectors with short ground returns.
F3 -> J11, F4 -> J12: independent PTC output rails. J10 carries no power:
the kicker board is supplied by the powerboard.
MF~MSMF075: each holds 0.55 A at 50C; NOT a combined available budget.
Shared U8/source limits still apply: fault selectivity is not guaranteed.
Shared I2C signals remain vulnerable to any device holding a bus line low.

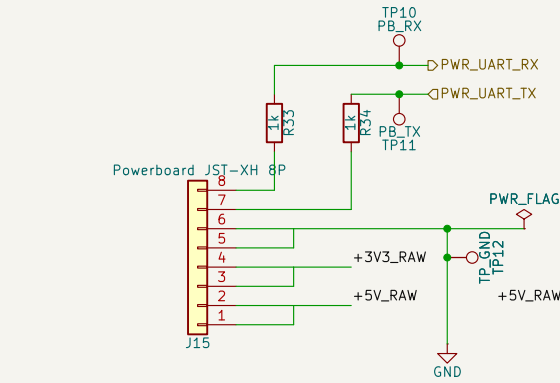


5V POWER PATHS: J14 INPUT LINKS / J16 LOAD SELECT

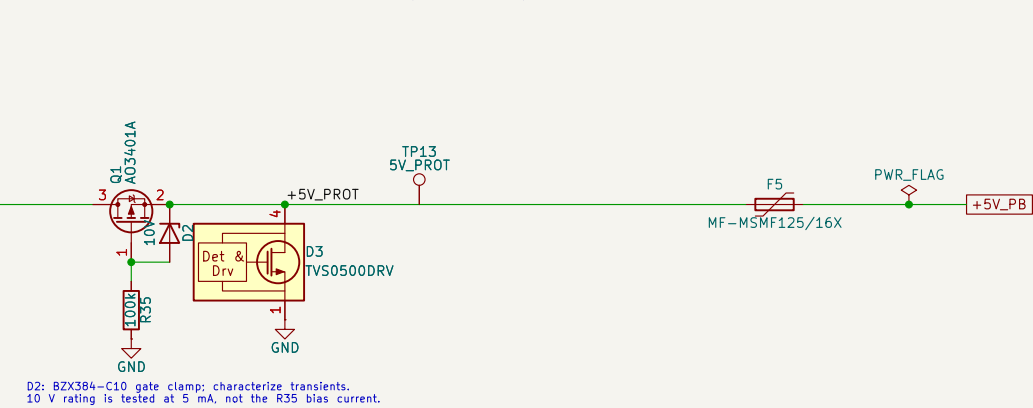
3V3 POWER PATHS: J17 INPUT LINKS / J18 LOAD SELECT



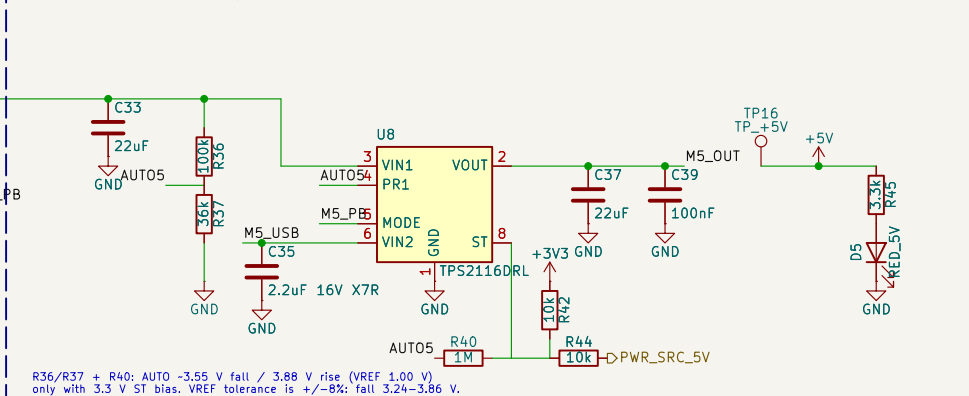
POWERBOARD CONNECTOR J15



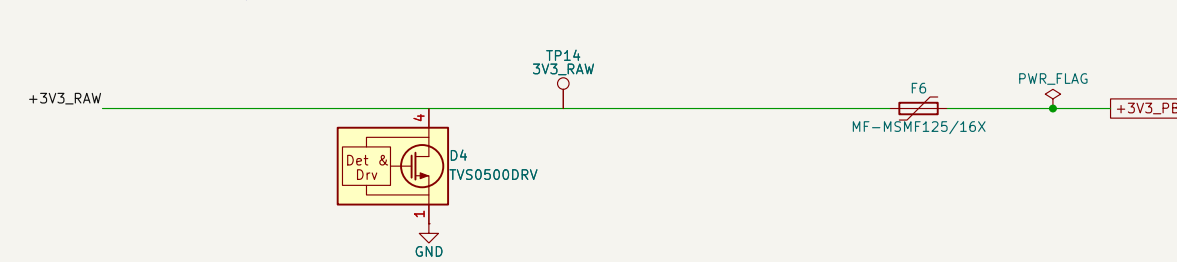
5V INPUT: Q1 REVERSE POLARITY, D3 TVS, F5 FUSE



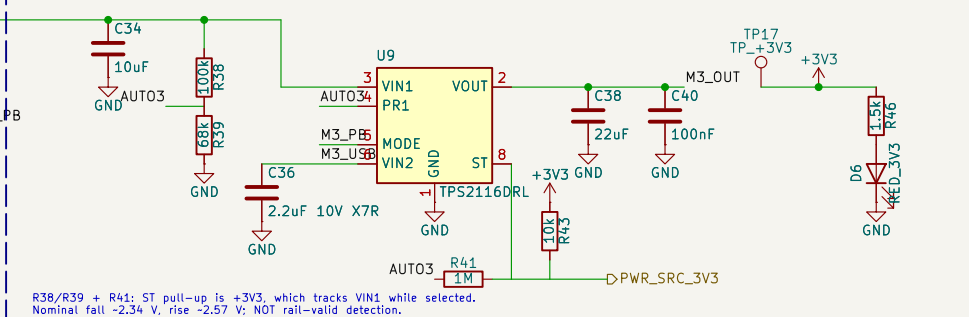
5V MUX U8: AUTO; BYPASS AT J16



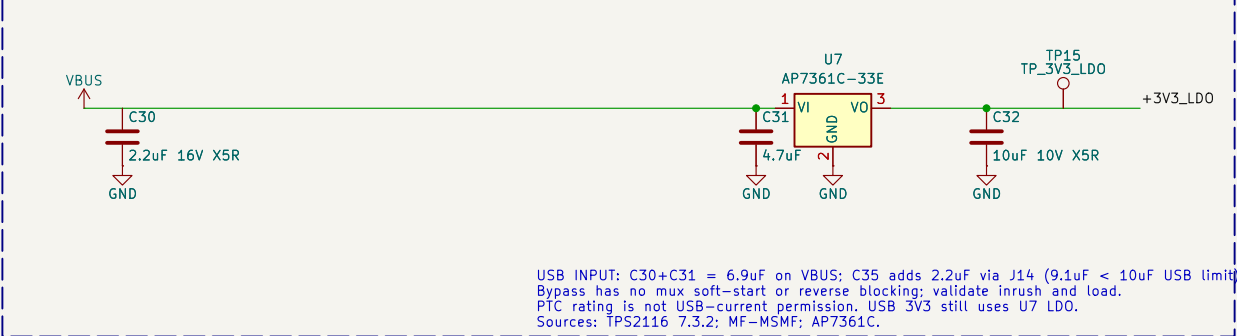
3V3 INPUT: D4 TVS, F6 FUSE



3V3 MUX U9: AUTO; BYPASS AT J18



USB VBUS AND 3V3 LDO U7



PROTECTION LIMITS: TVS0500 clamps up to 9.2 V at 43 A.
This is NOT proof of protection for 6 V-absolute-max TPS2116 or 3V3 loads.
PTCs are thermal devices, not fast current limits; verify fault coordination.
Sources: TVS0500, TPS2116 absolute ratings, Bourns MF-MSMF.

PHYSICAL BYPASS: J16/J18 carry load current, not PR1 control.
For recovery: power OFF, remove all J14/J17 caps, select PB or USB on J16/J18.
Bypass omits mux reverse blocking / soft-start. Source fuses remain.
ST reports MUX state only; cannot identify bypass source or USB current budget.
Control/status resistors remain; NOT galvanic or arbitrary-fault isolation.
Sources: TPS2116 SLVSFG1A 7.3 / 7.6.1; Samtec TSW and SNT datasheets.

J15: 1-2 +5V; 3-4 +3V3; 5-6 GND; 7 TX; 8 RX.
Mate/cable pinout and load remain external-interface requirements.
Q1 polarity agrees with AO3401A pinout. No 3V3 reverse-polarity FET.
R33/R34 limit current; they do not guarantee powered-off isolation.
PTC faults and mux source change can still interrupt other loads.

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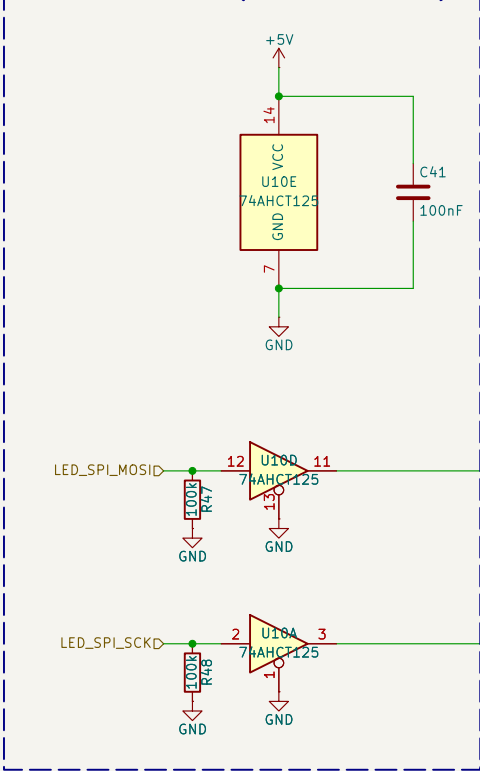
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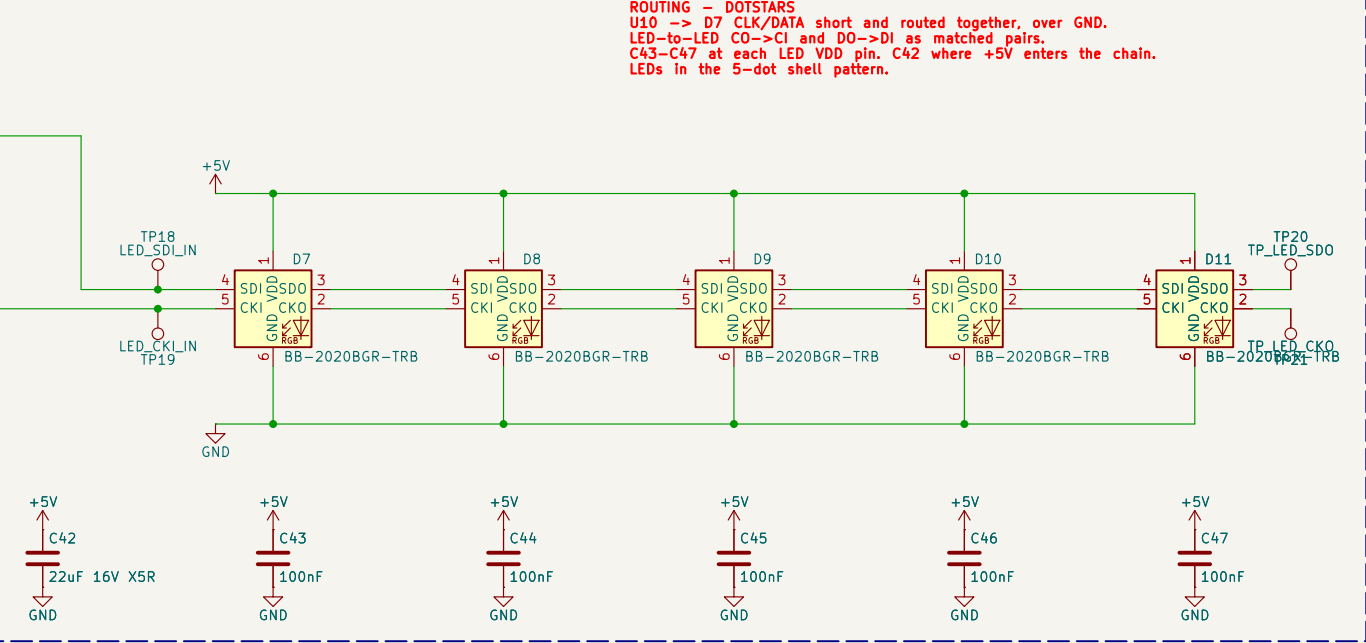
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Id: 6/10

LEVEL SHIFT U10 (3.3V SPI TO 5V)

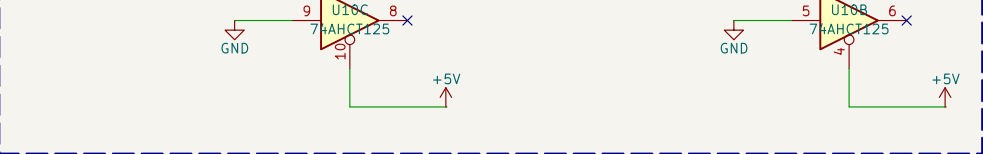


DOTSTAR CHAIN D7-D11 AND DECOUPLING



ROUTING - DOTSTARS
U10 -> D7 CLK/DATA short and routed together, over GND.
LED-to-LED CO->CI and DO->DI as matched pairs.
C43-C47 at each LED VDD pin. C42 where +5V enters the chain.
LEDs in the 5-dot shell pattern.

U10 SPARE GATES



LED LEVEL SHIFT: U10 = SN74AHCT125, specified for VCC 4.5-5.5 V.
3.3 V logic satisfies TTL input levels at a valid U10 supply.
R47/R48 = 100k bias inputs low; unused gates disabled.
D7-D11 = American Bright BB-2020BGR-TRB; use its data sheet/pad drawing.
A muxed USB source does NOT guarantee >=4.5 V after all cable/fuse drops.
Firmware current caps are requirements, not evidence of implemented limits.
Verify LED startup/current, power budget and custom footprints.
TP18/19 check chain input; TP20/21 check chain output.

Sheet: /DotStar LEDs/
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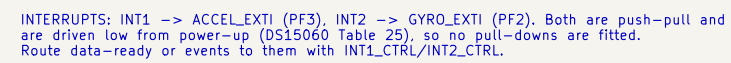
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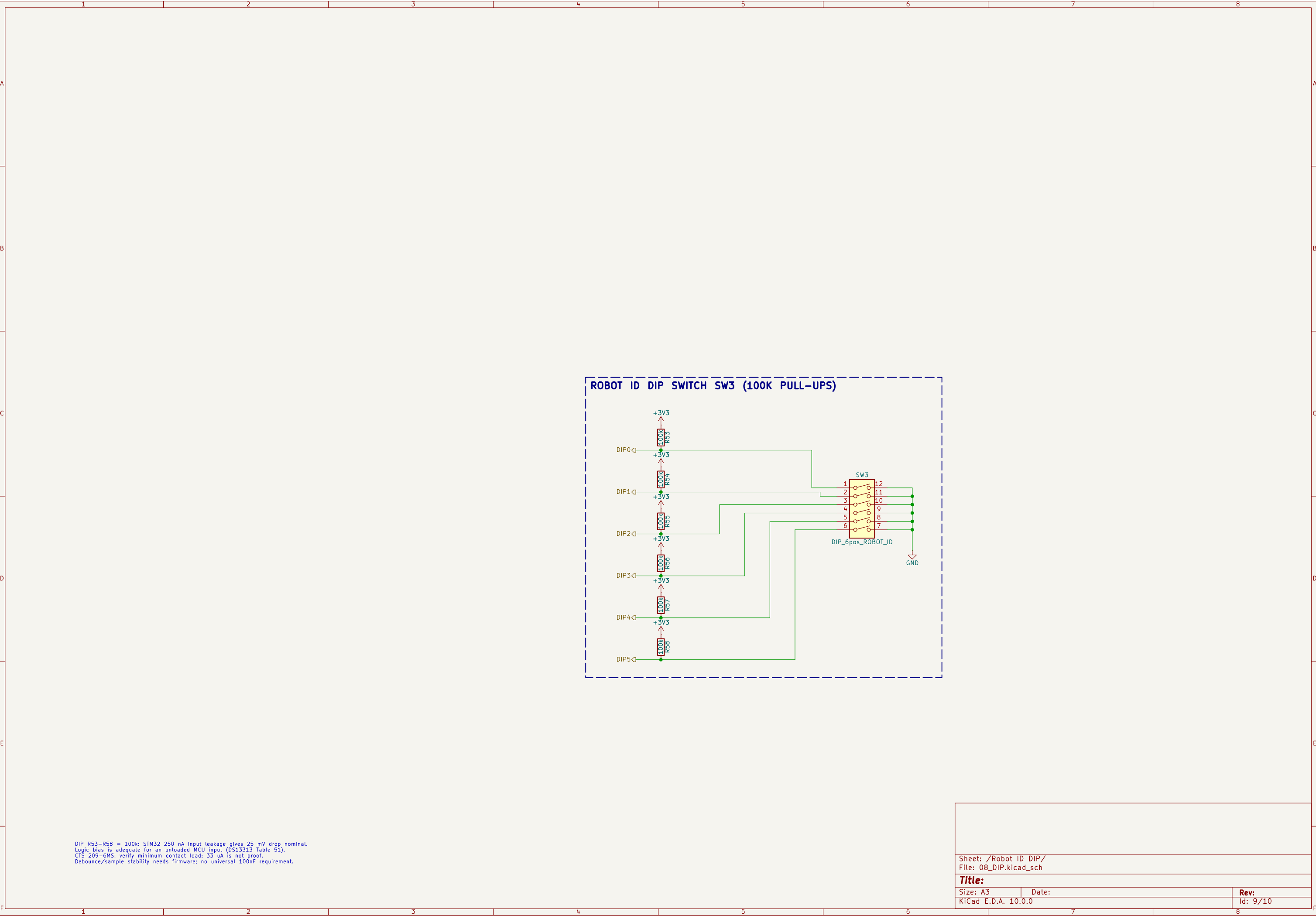
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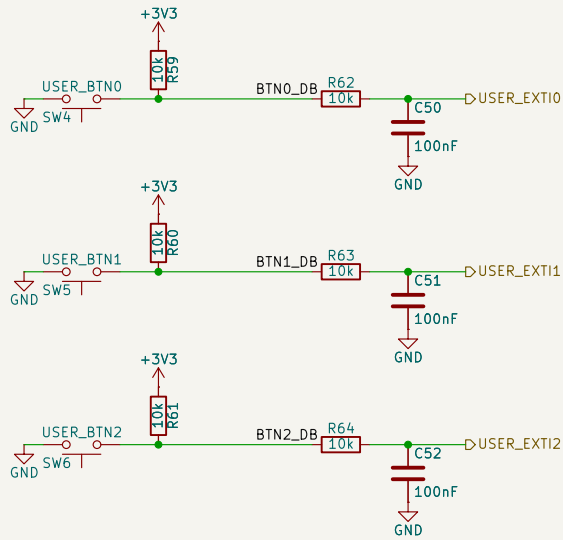
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USER BUTTONS SW4-SW6 WITH RC DEBOUNCE



BUTTONS: R59-R61 pull up with 10k; R62-R64 series 10k; C50-C52 = 100nF.
Calculated nominal time constants: press 1 ms, release 2 ms.
At +/-1% R and +/-10% C: 0.891-1.111 ms and 1.782-2.222 ms.
BOM: R59-R64 are 1%; C50-C52 need a +/-10% X7R part.
RC filtering does not guarantee removal of all bounce; quality switch and firmware.
Series R is between switch node and MCU/cap node, not between cap and MCU.
Input leakage check uses STM32H723 DS13313 Table 51.